Within the flat 6D Conformal Spacetime Algebra system ($Cl(4,1,1)$), the 2N2907 Bipolar Junction Transistor (BJT) can be modeled as the complementary counterpart to the NPN 2N2222 [Section 19].

Because the 2N2907 is a **PNP** silicon planar epitaxial switching transistor, its current conduction is driven primarily by the diffusion, recombination, and electric-field sweeping of positive core vacancies (holes [5.2, 5.7]) through a narrow, electron-soliton-doped Base region [4.1].

Section 20: The PNP Bipolar Junction Transistor (2N2907 Soliton Junction)

The 2N2907 consists of three complementary semiconductor regions: the heavily doped P-type Emitter, the narrow N-type Base, and the moderately doped P-type Collector.

```
        [ Emitter (P-type) ]   [ Base (N-type) ]   [ Collector (P-type) ]
        (High Density of       (Narrow Region of   (Moderately Doped with
         Positive Vacancies)    Electron Solitons)  Positive Vacancies)
                |                          |                        |
  Terminals:  Emitter (E)              Base (B)               Collector (C)
```

20.1 Physical Architecture and complementary Soliton Profiles In this system, the physical regions of the 2N2907 are modeled by their specific complementary soliton distributions: 1. **Emitter (P-type):** A heavily doped semiconductor region with a high spatial density of localized positive core vacancies (holes [5.2]). 2. **Base (N-type):** A narrow region ($W_B \approx 0.5\ \mu$m) containing mobile electron solitons (loops [4.1]). 3. **Collector (P-type):** A moderately doped region containing positive core vacancies, designed to absorb the vacancy flow from the Base.

20.2 Base-Emitter (BE) Junction and Vacancy Injection Applying a forward bias potential $V_{EB} > 0$ (Emitter positive relative to Base) lowers the electrostatic barrier across the BE junction depletion region of width $d_{\mathrm{BE}} \sim \delta_{\mathrm{BE}}$ [Section 3].

Under the inhomogeneous field equation $D \cdot F_{\mathrm{total}} = J_{\mathrm{total}}$ [2.4], this allows the heavily concentrated positive core vacancies in the Emitter to be injected across the junction into the Base. The injection current I_E is:

$$I_E \approx I_{ES} \left(e^{\frac{eV_{EB}}{k_B T}} - 1 \right)$$

where I_{ES} is the reverse saturation current of the BE junction.

20.3 Base Transport, Recombination, and Collector Sweep Once injected into the Base, the positive vacancies act as minority carriers within the electron-soliton-doped region.

20.3.1 Newtonian Diffusion Because the density of positive vacancies is high at the BE boundary and zero at the collector-base (BC) boundary, a strong spatial concentration gradient drives a classical, one-dimensional diffusion of positive vacancies across the Base width W_B.

20.3.2 Recombination (Base Current I_B) As the positive vacancies diffuse through the Base, a small fraction of them collide with and recombine with the Base's mobile electron solitons [4.1, 5.7]. This process is driven by the Base current I_B, which supplies electron solitons to replace those lost to recombination:

$$I_B \approx \frac{Q_p}{\tau_n}$$

where: * Q_p is the total charge of the positive vacancies currently in the Base. * τ_n is the physical recombination lifetime of the positive vacancies in the base lattice.

20.3.3 Collector Sweep (Collector Current I_C) The collector-base (BC) junction is reverse-biased ($V_{BC} > 0$, Base positive relative to Collector), generating a strong static electric field E_{BC} pointing from Base to Collector.

Because the physical width of the Base is narrow ($W_B \ll L_p$), most of the diffusing positive vacancies reach the BC boundary before recombining. Upon reaching this boundary, they are swept across the BC depletion region into the Collector by the strong E_{BC} field, constituting the Collector current:

$$I_C \approx \frac{Q_p}{\tau_{\text{base}}}$$

where τ_{base} is the average transit time required for a positive vacancy to diffuse across the Base.

20.4 Derivation of the Current Gain (h_{FE}) The static current gain h_{FE} (or β) of the 2N2907 is the ratio of the Collector current to the Base current:

$$h_{FE} = \beta = \frac{I_C}{I_B} = \frac{\tau_n}{\tau_{\text{base}}}$$

The average transit time τ_{base} for a positive vacancy diffusing across the Base width W_B with diffusion coefficient D_p is:

$$\tau_{\text{base}} = \frac{W_B^2}{2D_p}$$

Substituting this into the gain equation:

$$h_{FE} = \frac{2D_p\tau_n}{W_B^2}$$

The Mobility Constraint: Positive vacancy (hole) mobility in silicon is lower than electron soliton mobility ($\mu_p \approx 450\ \text{cm}^2/\text{V}\cdot\text{s}$ vs $\mu_n \approx 1400\ \text{cm}^2/\text{V}\cdot\text{s}$ [5.7]). By Einstein's relation ($D = \mu k_B T/e$), the diffusion coefficient of the positive vacancies is also lower ($D_p < D_n$).

To achieve a comparable current gain ($h_{FE} \approx 100$ to 300) under the same base width $W_B \approx 0.5\ \mu\text{m}$, the recombination lifetime of vacancies in the base (τ_n) must be longer, or the base doping profile must be adjusted to minimize recombination.

20.5 Transition Frequency (f_T) and Transport Limits The lower mobility of the positive core vacancies directly affects the transition frequency f_T:

$$f_T = \frac{1}{2\pi\tau_{EC}}$$

where τ_{EC} is the total emitter-to-collector delay time:

$$\tau_{EC} = \tau_E + \tau_{\text{base}} + \tau_C + \tau_D$$

For the 2N2907, the minimum transition frequency is:

$$f_T \approx 200\ \text{MHz}$$

This is lower than the 2N2222's 300 MHz, corresponding to a longer total emitter-to-collector transit delay of:

$$\tau_{EC} = \frac{1}{2\pi \times (200 \times 10^6\ \text{s}^{-1})} \approx 0.80\ \text{nanoseconds}$$

This 0.80 ns delay represents the physical, deterministic limit of the speed at which the positive vacancies can diffuse and drift through the complementary semiconductor crystal of the 2N2907, defining its switching and amplification boundaries.